



**NET
NOD**

Survival in today's new geopolitical situation

Patrik Fältström · Netnod
NONOG, 9 September 2026

Where do I come from?

- Netnod – operator-neutral, nearly 30 years at the core of the Internet
- The largest Internet Exchange (IX) in the Nordics
- i-root – one of the world's thirteen DNS root name servers
- Swedish time and frequency is distributed by Netnod on behalf of the Swedish Post and Telecom Authority
- Various security services
- Netnod and NIX have worked closely together for 20 years to help build resilience in the Nordics



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Stochastic failure – or an adversary?

BEFORE – chance

- Natural events, components failing, suppliers going out of business
- The event doesn't care who you are
- History predicts the future: MTBF, statistics, due diligence
- Two different solutions halve the risk
- Occurs stochastically, at any time

NOW – someone who chooses

- Russia, China and Iran. Direct attacks, often deniable via proxies.
- The adversary chooses you, based on your value – not on chance
- History provides little information. The process has changed, but analysing the adversary helps.
- Two solutions of a similar type provide no protection
- Occurs when and where you are at your weakest

Chance doesn't care who you are. An adversary does.

The risk matrix is built for the wrong world

BEFORE

- Low probability × high impact = accept
- It was a rational decision, easy to disregard
- Probability was drawn from history and experience
- In particular, risk related to infrastructure investments can be disregarded

NOW

- Probability has increased so much that the risk can no longer be disregarded
- Probability is a property of the adversary's intent – and that intent has changed
- Swedish Military Intelligence and Security Service (MUST): Russian activity against Europe is likely to increase in frequency, scope and risk-taking through 2030
- E-tjenesten (Fokus 2026): If security policy relations with Europe were to deteriorate further, Russia would be likely to carry out more frequent and more serious operations.

Accepting a risk was the right decision. Not revisiting that decision is not.

Four different attack vectors

Direct — the state actor itself

- Intelligence activity, cyberattacks, sabotage

Via proxy — commissioned criminal networks

- State actors use criminal networks and disposable agents as deniable proxies

Deniable — covert influence activities

- Gets others to act with no trace back to the party behind it

Information campaigns — control the adversary

- If you behave as though you've been attacked, it has worked.

**The incident looks like a crime.
The adversary can be someone else.**



Via proxy — commissioned criminal networks

How it works

- A third party acts for a state — deniability for the state, risk for the proxy

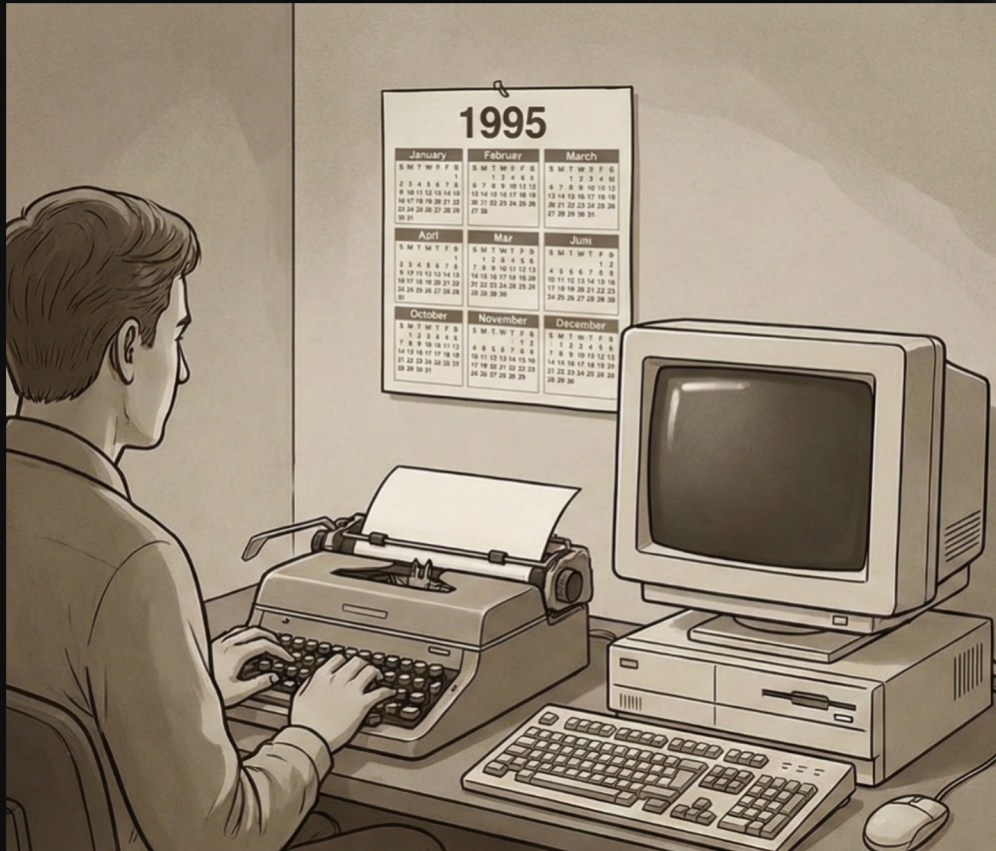
Sweden

- SÄPO: Iran uses Swedish criminal networks as a proxy
- Russia recruits disposable agents for one-off attacks

Norway

- Server Killers (pro-Russian) declared 'cyberwar' against Norway, 26 August 2026
- Trigger: drone support deal with Ukraine, 23 August — Digdir hit the next night
- ID-porten, Altinn, e-Innsyn + 7 other services disrupted

The incident looks like a crime. The adversary can be someone else.



Before: Digital complements



Now: Digital dependencies

Effect

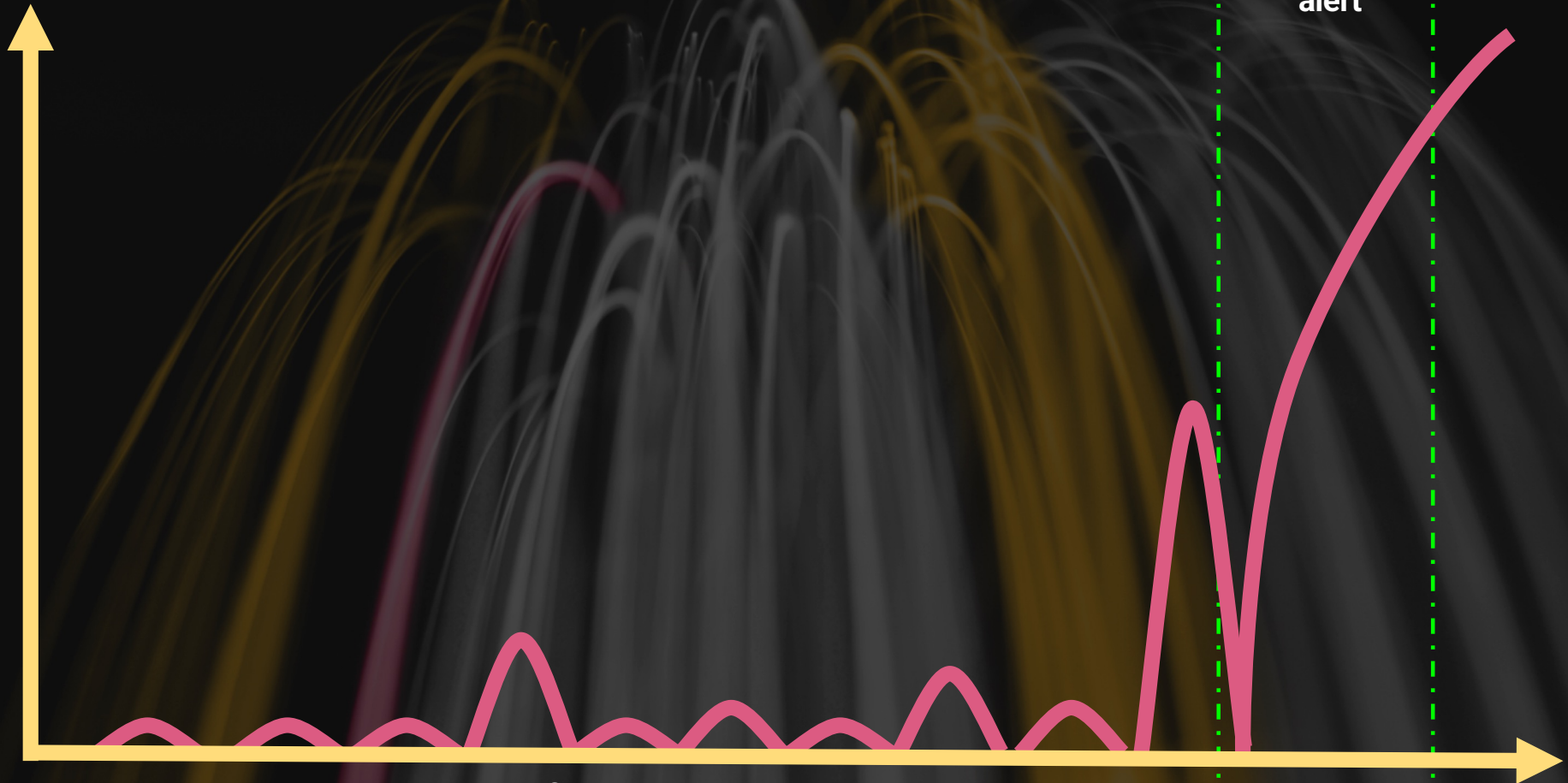
Peace

Heightened
alert

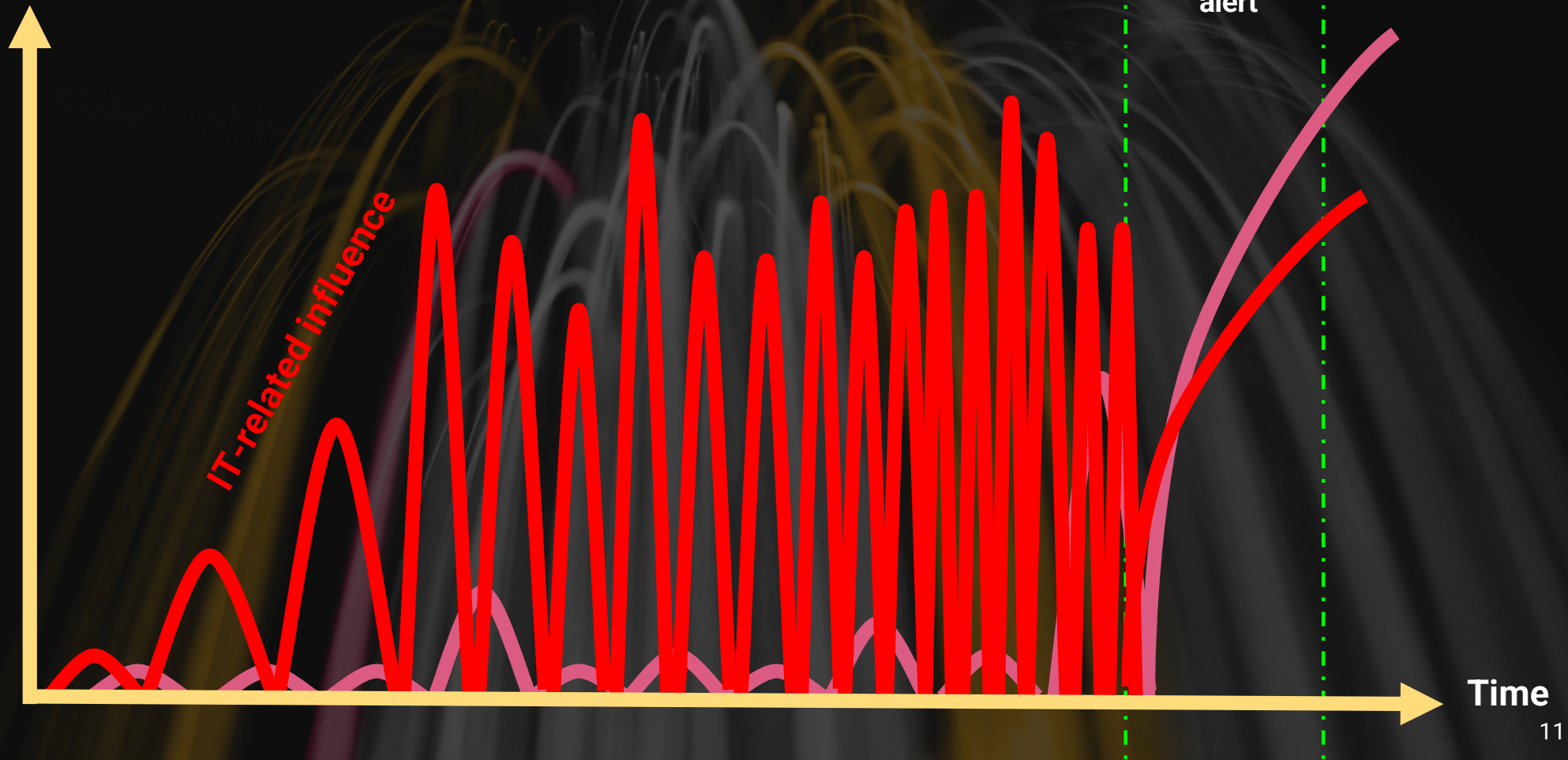
War

Traditional influence

Time



Effect





**“Everyone thinks they have a plan
until they get punched in the mouth”**

– Mike Tyson 12

AI compresses the timeline

The attack chain gets shorter, not different

- Recon, phishing, vulnerability analysis, lateral movement – faster and in parallel

Response times assume a timeline that no longer exists

- Alarm at night, on-call called in, decision at the morning meeting, action after lunch
- Attacker works in minutes, you work in business days – the plan is irrelevant

Consequence for risk analysis

- RTO/RPO are set against the wrong clock

We must work faster than the enemy. Slowness becomes reactive work.

Agents are insiders

Permissions define the attack surface, not model capability

- An agent with credentials and tool access acts across system boundaries
- Do you know how many non-human identities you have – and what they may do?

An agent's work crosses trust boundaries

- Instructions, retrieved content, tools and memory – not just the prompt
- The risk isn't that work happens; it's the action it can trigger

Questions to ask

- How large is the attack surface (permissions)? How easily are boundaries crossed (architecture)?

Have you granted permissions to something you don't control?

Regulatory developments

New and updated regulations related to cybersecurity

EU

Digging into ex-ante regulation tied to a complex layer of processes

- CRA, NIS2, CER, DNA, etc.
- **ProtectEU:** Currently an unclear strategy for internal security. Affects electronic communications and operator liability.
- **Regulatory compliance?**

USA

Shifted approach to compliance during 2024 (Sept, DOJ ECCP)

- From "you must do it this way!" to "does what you're doing actually work?"
- *"Is the program well-designed? Is it adequately resourced and empowered? Does it work in practice?"*
- **Survival?**

This is not the future

The Cybersecurity Act (2025:1506) – in force since 15 January 2026

- Sweden's implementation of NIS2. In effect for seven months.

The CER Act – in force 1 January 2027

- Resilience of critical entities, Government Bill 2025/26:303. In four months.

General election in Sweden: 13 September 2026 – in four days

Sovereignty is an interesting word

Result of a measurement

europa.eu – sets the Union's assurance levels

Summering

6 länder

12 externa aktörer/domäner



BE · GB · IE · LU · MT · US

europa.eu – web hosted on own servers in Luxembourg (LU), DNS delegated to bt.net and cw.net (US-owned operators), email via Microsoft 365 (US) and Proofpoint (US), certificate from GlobalSign (BE), no DMARC

regeringen.se – published the cloud policy

Summering

9 länder

23 externa aktörer/domäner



BE · CA · CR · FR · GB · LU · NL · SE · US

regeringen.se – DNS on Swedish Excedo (SE, one nameserver in CA), web on Cloudflare (US), certificate from DigiCert (US), no CAA, page loads video from qbrick.com (FR) and text-to-speech from readspeaker.com (LU), DMARC reports via Amazon eu-west-1 (US), TXT-verified by Google (US)

An origin requirement doesn't measure this. A dependency analysis does.

Sovereignty is an interesting word

Result of a measurement

regjeringen.no – the Norwegian government

Summering

3 länder

10 externa aktörer/domäner



GB · SE · US

regjeringen.no – DNS on Azure (Microsoft, US), apex IP hosted in Sweden (SE, no IPv6), www on Cloudflare (US), certificate from Google Trust Services (US), no CAA record, DMARC reports to Amazon us-west-2 (US)

nsm.no – Nasjonal sikkerhetsmyndighet

Summering

7 länder

24 externa aktörer/domäner



AT · CR · FR · GB · NL · NO · US

nsm.no – DNS on Cloudflare (US), certificate from Google Trust Services (US), TXT-verified by Apple, Google and Microsoft 365 (all US), page CSP allows SoundCloud (FR) and Amazon (US)

An origin requirement doesn't measure this. A dependency analysis does.

Cloud and AI Development Act — COM(2026) 502

Tech Sovereignty Package, 3 June 2026 — targets non-EU hyperscalers controlling 70%+ of the EU cloud market.

- Swedish Government Offices reference number Fi2026/01676
- The consultation response is due by **2 October 2026**

What Netnod said:

- Requirements should be based on capability, not on where the supplier is established
- What needs protecting is the function that must be maintained
- Continuity should be verified, not just documented
- Non-EU suppliers must stay permitted — diversity is the point.

<https://netnod.se/cybersecurity/cloud-and-ai-regulation-must-be-built-on-capability>

Regulation must be built on capability — not on origin.

Norway and the EEA – an open question

The open question

- Does CADA treat Norway as a third country – or does EEA incorporation grant EU-equivalent status? The proposal does not say.

Why it matters in this room

- The proposal does not settle it – EEA relevance is decided in the incorporation process
- Until then, neither Norwegian providers nor Norwegian buyers of EU cloud have a clear answer

How to avoid the question entirely

- Judge capability and verified continuity – not country of establishment.

An origin test raises this question. A capability test never does.

The duality of security



Opposing architectures

Availability and confidentiality (often) require completely different technical solutions.

What are the actual requirements?



Availability

Best achieved through redundancy and diversity.

Many, varied nodes. High diversity.



Confidentiality

Achieved by consolidating and protecting information.

Few nodes, controlled design.



What are the real consequences?

Conflicting needs create a complex challenge for robust IT design.

Resilience in practice



Planning & Schedules

Creating and printing schedules for the next day may be doable; but can you do it for two weeks?



Hardware replacement

Central hardware isn't replaced in a day.



Frequency at which services are needed

Payroll is only needed once a month. Often the same pay can be issued for several months in a row.



Critical questions

What must work all the time? And what only sometimes?

IT does IT, not what the business needs



What needs have been communicated?

Does the business determine the requirements of the IT operation?



What mandates exist?

*Does the IT department know if a system is critical? Is there a mandate and resources for resilience?
Is a system allowed to be shut down or isolated?*



Example – close to home

No requirements were set for mobile phones, so the IT department bought the cheapest (or the fanciest).



Example – far away

NIS2 is delegated to the IT manager; the rest of the business carries on as usual.

Speed of action is a question of mandate, not technology.

Focus on confidentiality, not availability



Focus on data leakage over operational reliability

- What's worse: records leaking, or the ER shutting down?
- Technical decisions are made unilaterally out of fear of GDPR violations



Real-world example 1

Backup routines aren't tested because of fear that sensitive information might go astray during the test.

Focus on confidentiality, not availability



Focus on data leakage over operational reliability

- What's worse: records leaking, or the ER shutting down?
- Technical decisions are made unilaterally out of fear of GDPR violations



Real-world example 2

Because of GDPR, an application is run on in-house hardware with its own storage system. The storage system is hacked, and all the information is leaked.



Ask yourselves: do you know what you do, and what expectations exist?

- Do you know which IT systems and functions you need to keep doing what's expected of you for at least two weeks?
- Does the legislator know better than you what it takes to keep your organisation and its systems running?

In Sweden, the Swedish Civil Defence and Resilience Agency (MCF) recommends that households be able to cope for a week.

Can your organisation cope for a week?

Compliance is one thing, survival is another

The best way forward



1. Agree on

What you do, and what really matters



3. Ensure operations

Verify that critical functions work when needed



2. Map dependencies

Identify the vital functions and dependencies



4. Failure test

Disconnect a real supplier and verify continuity

The best way forward



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Disconnect a real



We need to work together

Nothing you do can be handled in isolation

- There are always dependency chains; you're just one link among many

What you do can be indirectly affected by events that hit others

- Even if you contractually require certain functionality, things can still go wrong

A chain is no stronger than its weakest link

- But with several chains, things will hold together until the strongest (last) chain breaks

Your own assessments in your risk analysis aren't enough

- You depend on the outside world, and the outside world depends on you

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